Artificial Intelligence

Use of Artificial Intelligence to Detect Unauthorized Access in Cloud File Storage Systems

Summary

Introduction
Unauthorized access remains one of the most serious threats to cloud storage systems. Traditional security mechanisms such as static access control and rule-based intrusion detection systems often fail to detect sophisticated and evolving attacks. This research explores the use of artificial intelligence techniques to enhance the detection of unauthorized access in cloud storage environments.

Goal of the Research
The goal of the research is to design and evaluate an artificial intelligence-based intrusion detection system capable of identifying abnormal and malicious activities in cloud storage systems. The study aims to improve detection accuracy while reducing false alarms.

Methodology
The researchers proposed a machine learning-based intrusion detection model trained on cloud security datasets containing normal and attack behaviors. Data preprocessing and feature selection techniques were applied before training the model. The proposed algorithm was compared with existing machine learning classifiers to assess performance improvements.

Results
The results show that the AI-based model achieved higher detection accuracy and lower false-positive rates compared to traditional approaches. The system effectively identified unauthorized access attempts by analyzing patterns in user behavior and network traffic. This demonstrates the capability of AI techniques to adapt to dynamic cloud environments.

Conclusion
The study concludes that artificial intelligence provides a powerful solution for detecting unauthorized access in cloud file storage systems. AI-based intrusion detection systems enhance security by learning complex attack patterns and responding proactively. This approach is suitable for securing cloud-based file storage systems in educational institutions where data confidentiality is critical.

APA References

Securing the cloud storage using novel machine learning-based intrusion detection system. (2024). *International Journal of Intelligent Systems and Applications in Engineering*. https://ijisae.org/index.php/IJISAE/article/view/5614